Global Ocean Accounts Partnership

The Ocean Economy: Satellite Accounts

Understanding the ocean's economic contribution

Maria Alarcon

Lecture prepared by Dr Jordan Gacutan, adapted from Prof. Charlie Colgan (Middlebury Institute)

GOAP

Learning outcomes

1. Describe Ocean Economy Satellite Accounts (OESA) and their relationship with the national accounts,
2. Describe the general process of their compilation
3. Identify two examples of existing OESA

GOAP

What are Ocean Economy Satellite Accounts?

- Satellite Account focused on ocean-related economic activities
- Maintains consistency with the national accounts
- Identifies aspects that **aren't visible** in standard national account presentations

GOAP

Ocean Economy Satellite Accounts

By definition:

- **Shipping**
- **Fisheries**
- **Ports**

- **Tourism**
  - Coastal and marine tourism
- **Insurance**
  - Marine insurance
- **Construction**
  - Marine construction
  - Coastal engineering

Partially derived / dependent

GOAP

Why do we need Ocean Economy Satellite Accounts

Policy development and planning:

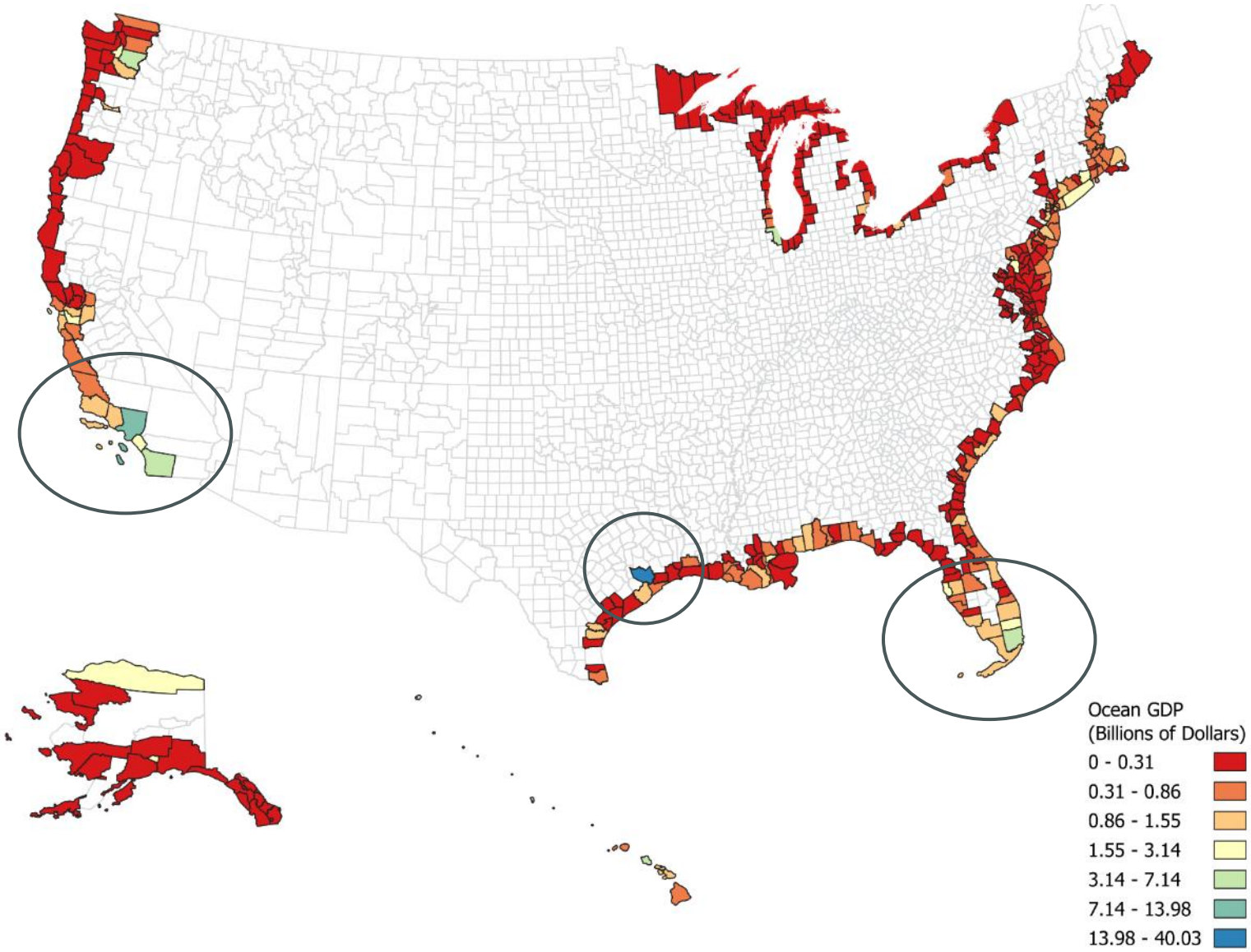
- Basis for strategic goals and policies for the ocean economy
- Produce indicators to measure the effectiveness of policies

Industry analysis:

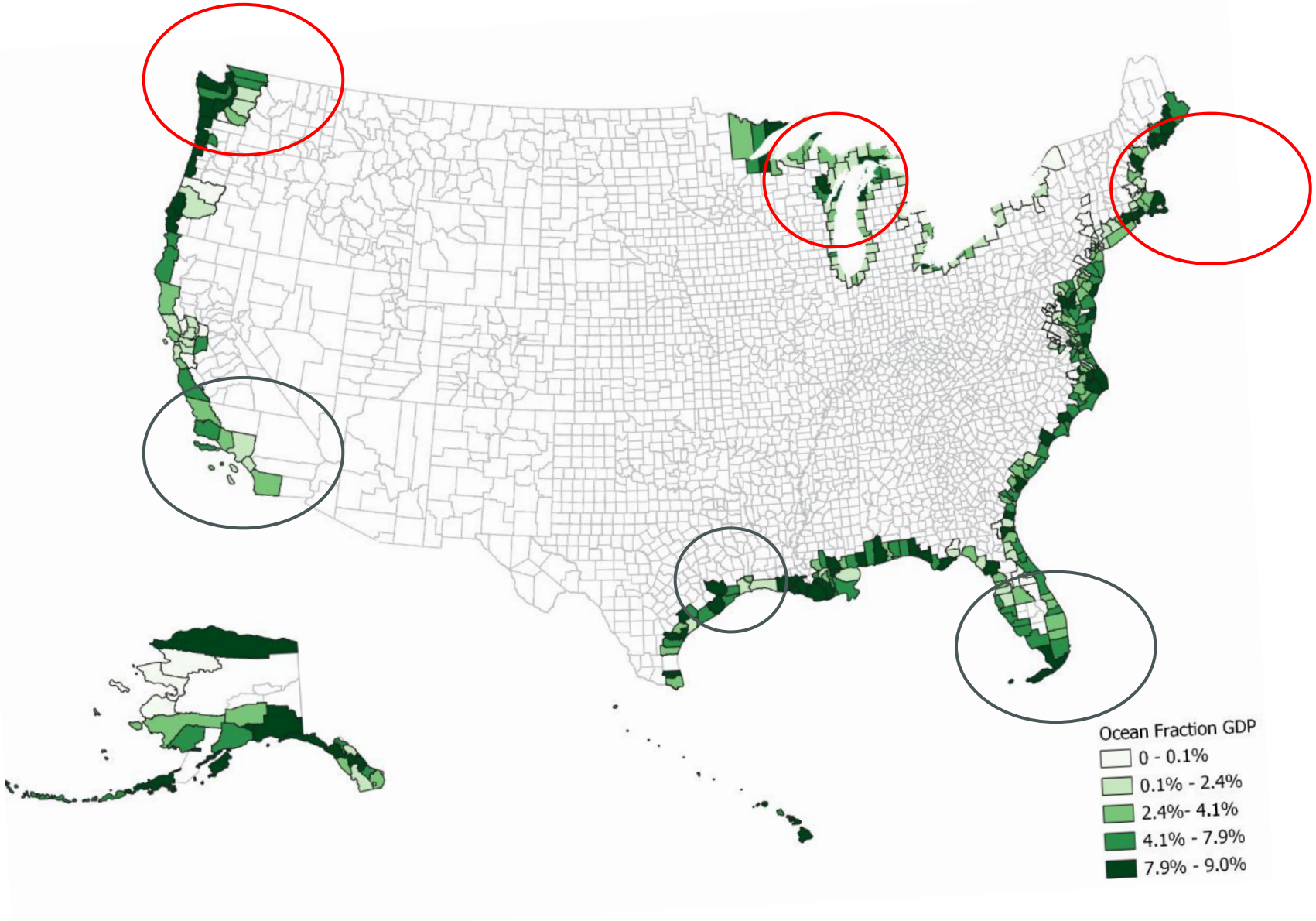
- Relationship between ocean economic sectors and interrelationships.
- How large is Belize's coastal tourism sector?
- What is the growth of key ocean sectors?

GOAP

Ocean GDP (billions of dollars)



Ocean GDP Share as Percent Total GDP



Overview

- Components of OESA
- GNP
- Geography
- Industry and partials

GOAP

1.

What portion of the

national (or regional) production
of goods and services

is related to the
ocean?

2.

3.

Component 2. Calculating national (or regional) production of goods and services

Note difference between "Domestic" and "National""

- Domestic: Production in Country X
- Production by Citizens of Country X

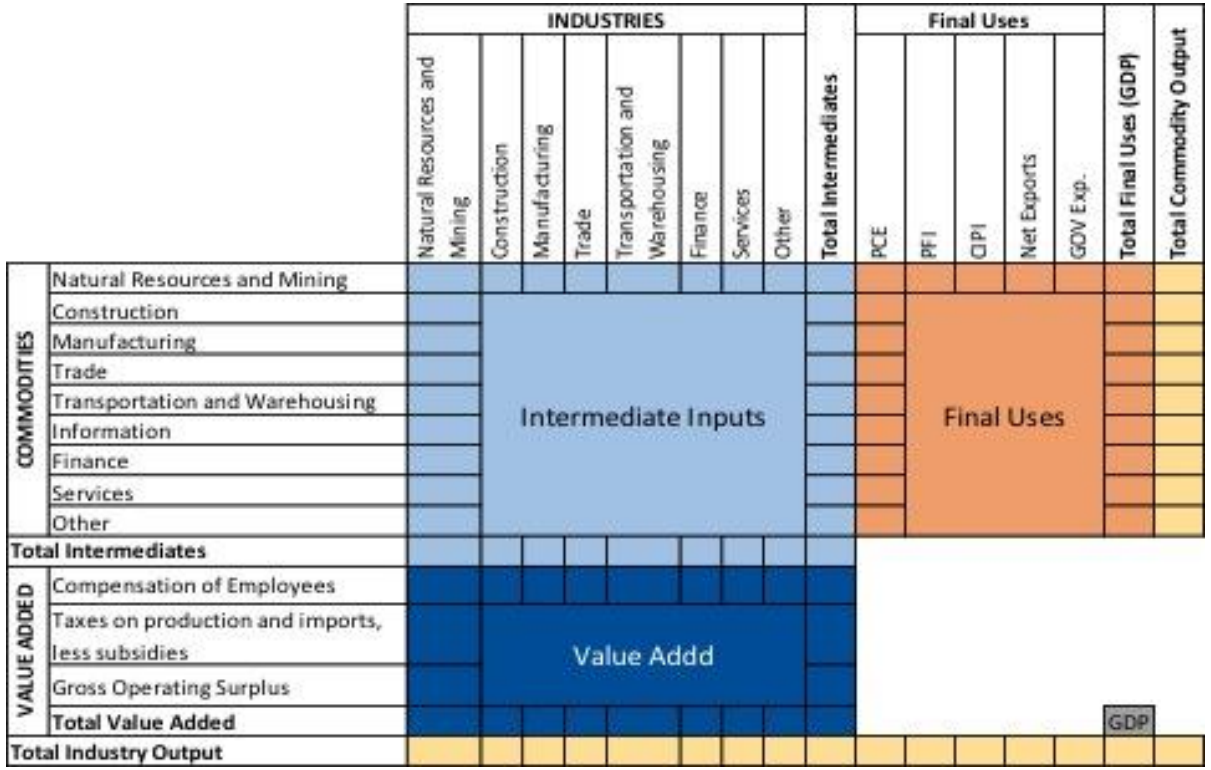
GNP = GDP + Net factor income from abroad

Where:
Net factor income from abroad =
Income received by resident units from abroad -
Income paid to non-resident units

Completing the Accounts: Input-Output Tables

Three general approaches (often all):

- Production / Value Added Approach
- Income Approach
- Expenditure Approach



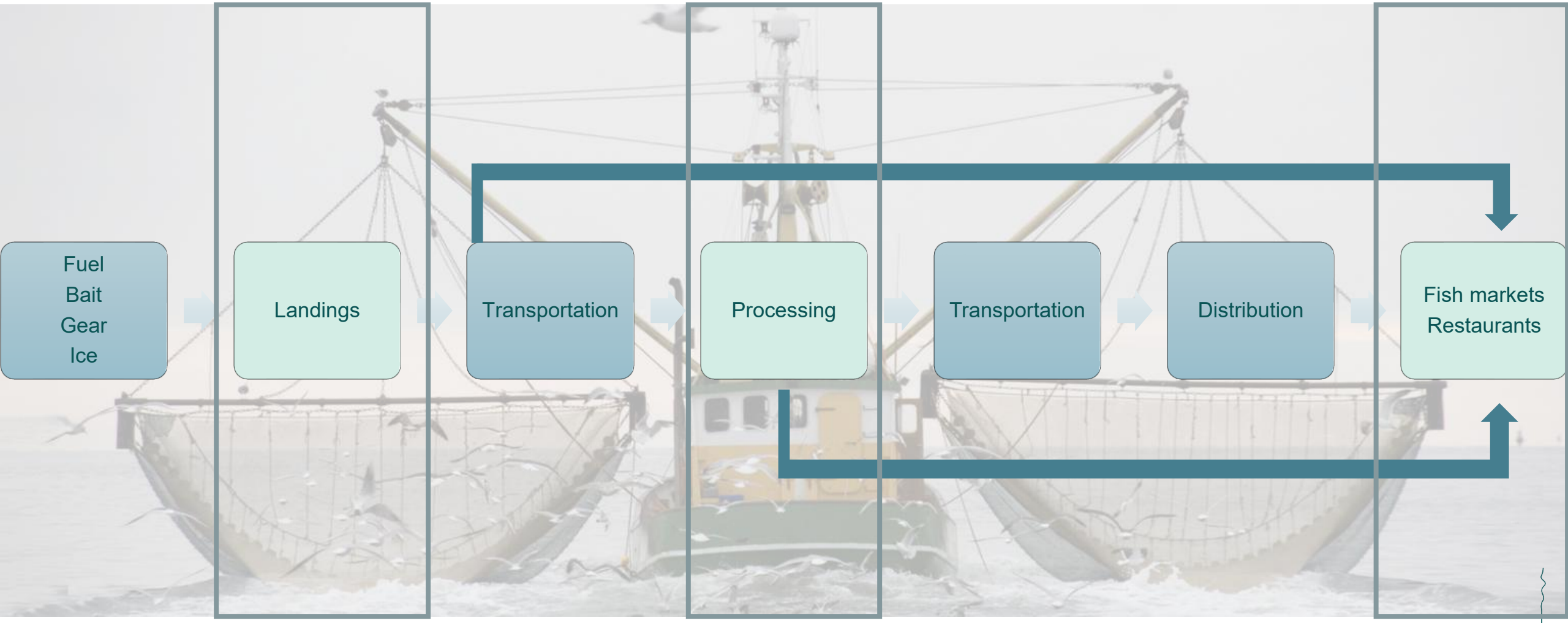
Why Value Added?

Gross Value Added (GVA) = Gross Output - Intermediate Inputs

GDP = Sum of GVA added across all industries (+ tax – subsidies)

1. Provides Output by Industry
2. Tracks the Value Chain
3. Can be measured in different geographies

Value Chain for Fishing



GOAP

1.

What portion of the

national (or regional) production
of goods and services

is related to the
ocean?

2.

3.

GOAP

Geography

National Accounts (SNA) → Ocean

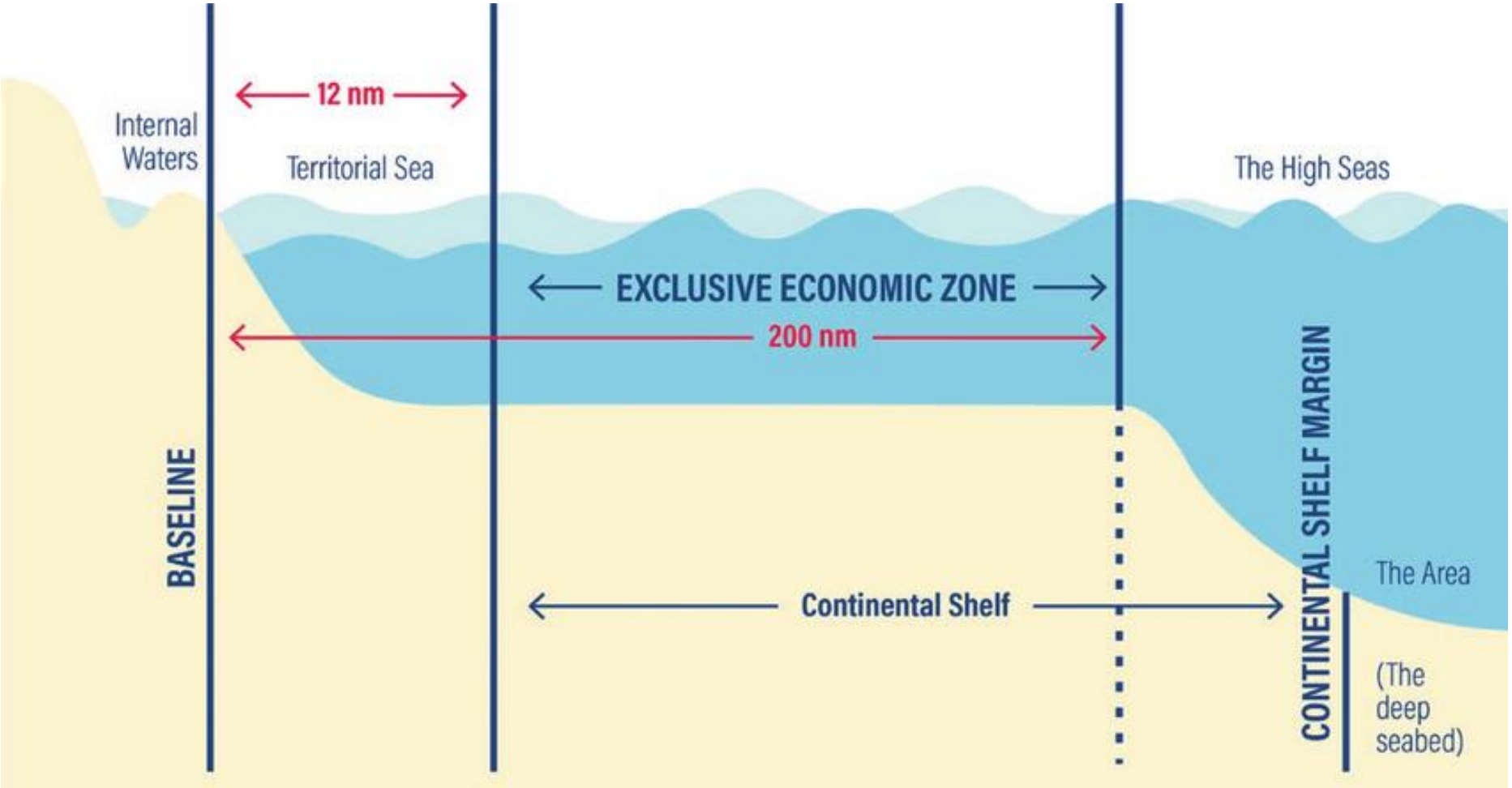
Statistical geography, defined by:

- Land and sea boundaries

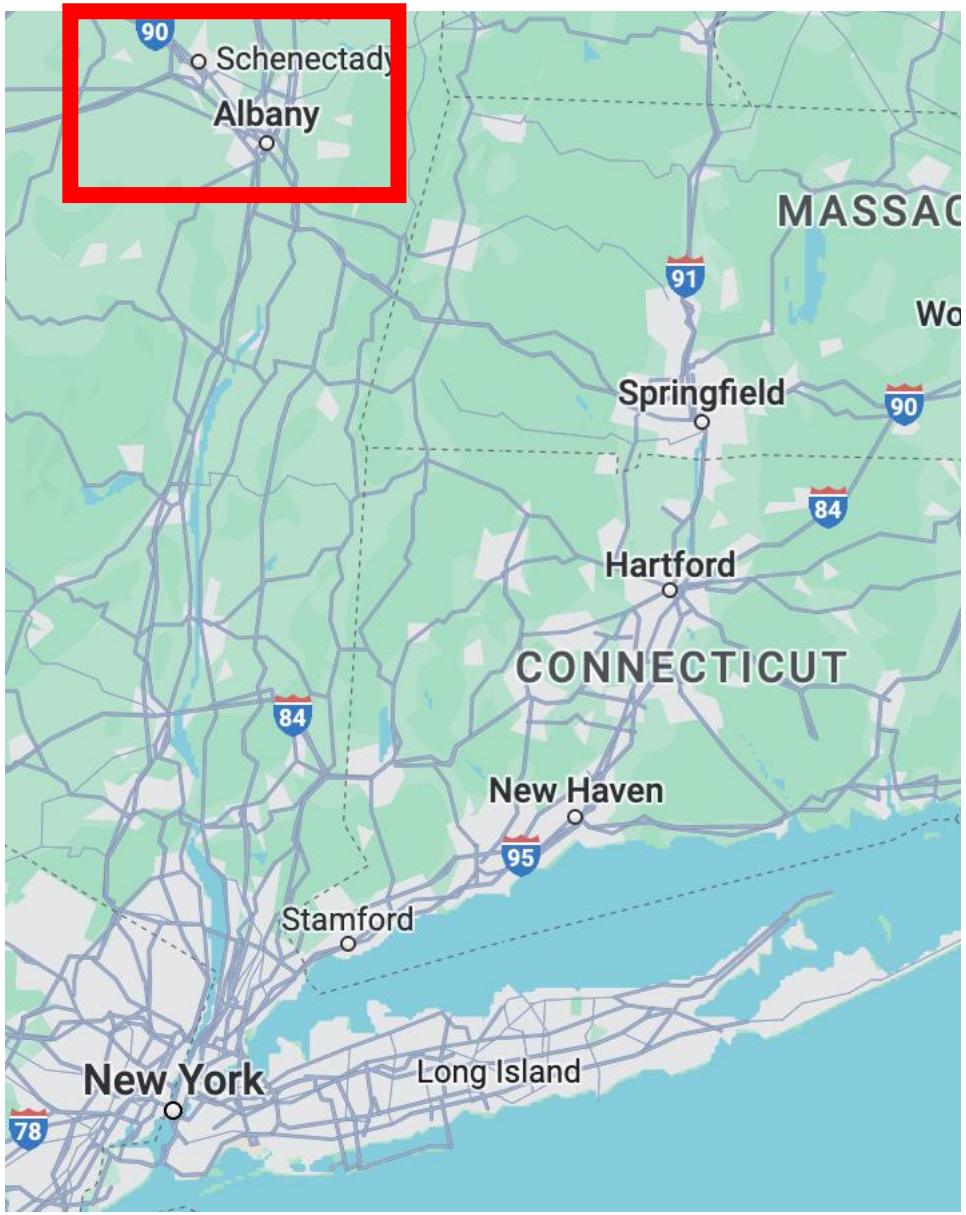
Q: What are examples of sea boundaries?

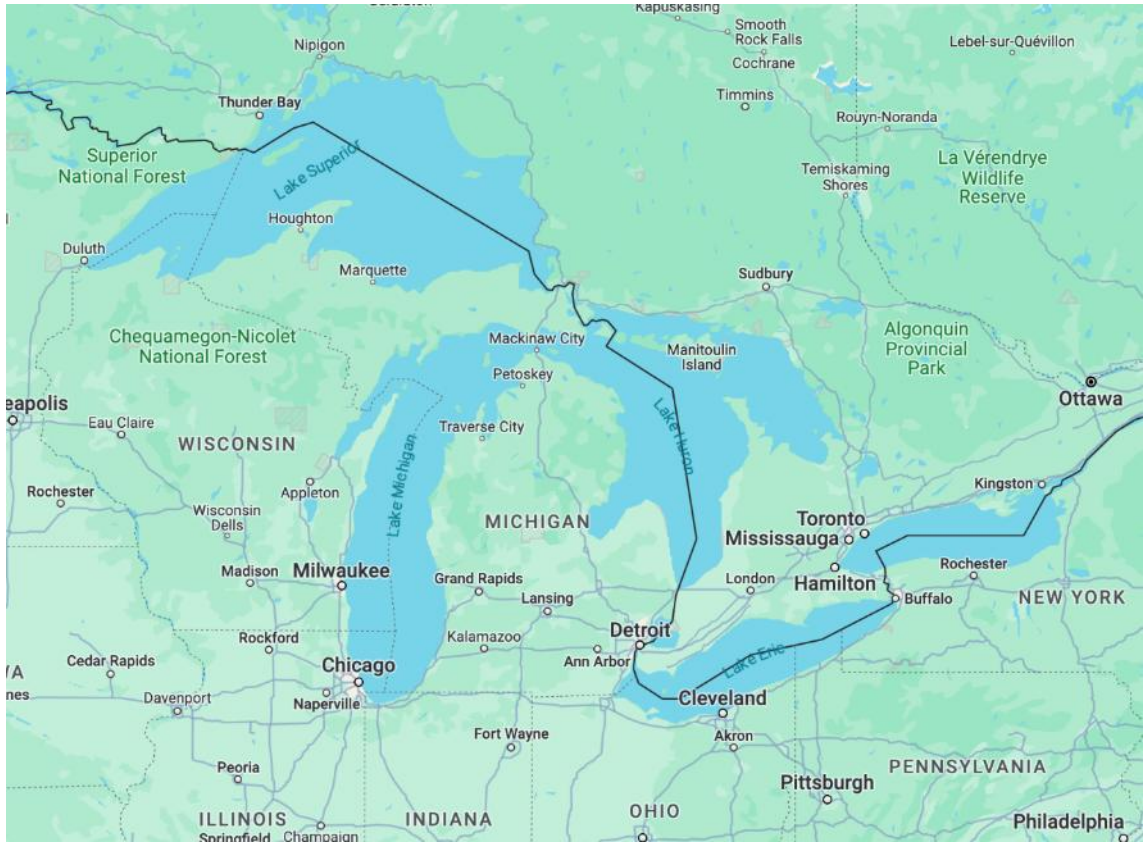
GOAP

Component 3. Ocean Law of the Sea – sea boundaries



Where does the Ocean stop?





USA – defines Ocean as 50 km inland and Great Lakes: 244,106 km² (SL = 62, 705 km²)

Where does the Ocean stop?

GOAP

Quick discussion

- What are the **geographic statistical areas** and their levels in Belize?
- Are there any laws or customs in Belize that define ocean limits **landwards**?

1.

What portion of the

national (or regional) production
of goods and services

is related to the
ocean?

2.

3.

GOAP

Estimate the “Ocean” portion of industries

- Option 1: Disaggregate from the national accounts (Ireland)

- Option 2: Aggregate up from individual records (USA, S. Korea)

Option 1: Disaggregation

National output to industry

Determine partial using data, assumptions and/or expert elicitation from industry.

Ocean component of Hotels and Lodging Places:

- Partial = Proportion of leisure-related travel in shore adjacent geographies (0.28 of consumer expenditure).
- Derived from final demand.

Trade off between precision with data requirements.

Methods for Option 1: Disaggregation

Method depends on data, time and resources available.

Where “partial” = ocean component of an industry or activity.

1. Values Added for Industry (i) * partial
   - Simple but ignores interdependices
   - Subject to double counting
2. Supply table entries * partial
3. Final demand by product category (CIGX) * partial
   - More complex and data intensive

GOAP

1: Value Added by Industry × Partial

- This is the **simplest** method.
- You take the **gross value added (GVA)** for an industry (e.g., all of manufacturing), and multiply it by a **partial** to get the ocean-related share (e.g., boat-building).

  📌 *Example: If the manufacturing sector's GVA is $10B and we estimate 5% is for marine vessels → Ocean share = $10B × 0.05 = $500M*

✅ Pros: Easy to do, quick to implement.

⚠️ Cons:

- **Doesn't account for interdependencies** between industries (e.g., how marine tourism drives other sectors like food or retail).
- **Can result in double counting** if multiple sectors are interlinked and all have partials applied separately.

2: **Supply Table Entries × Partial**

- Uses the **supply side of the economy** (i.e., which industries produce which goods/services).
- Here, you look at **products** or **commodities** being produced and determine which are ocean-related.
- Then you apply a **partial** to the total production of those goods.

  📌 *Example:*
  You look at the supply table and see how much "seafood" is being produced—if 90% of seafood is marine and 10% freshwater, you apply those partials.

✅ More accurate than Method 1 because it's product-specific.

⚠️ Requires **supply-use tables** and more detailed data—might not be available in all countries.

3: **Final Demand (C + I + G + X) by Product × Partial**

- This method works on the **demand side** of the economy:
  - **C** = Household consumption
  - **I** = Investment
  - **G** = Government spending
  - **X** = Exports
- You apply partials to the final demand for products/services to estimate what portion is related to ocean industries.

  📌 *Example:*
  If total household spending on travel is $2B, and 28% of that is for coastal tourism →
  Ocean tourism = $2B × 0.28 = $560M

- ✅ **Most accurate** when done well because it ties directly to consumer behavior and demand for marine products/services.
- ⚠️ **Most complex** and **data intensive**—requires disaggregated data on household spending, tourism, exports, etc.

GOAP

Option 2: Aggregation

Performed using existing data or projects within country, supplemented by Surveys

- Identify sample frame of Ocean-Related firms
- Period surveys that supplement existing data collection

Challenges:

Method significantly challenged by access to geolocated data. There are assumptions with using location and industry as a proxy for ocean relationship.

⚠️ Challenges

- **Access to geolocated data**
  - You often rely on **location** as a proxy (e.g., assuming firms near the coast are ocean-related), but that's not always accurate.

- **Assumptions about industry and location**
  - Some firms near the coast might not be marine (e.g., a bakery), and some marine businesses might not be right on the shoreline (e.g., a shipping logistics company inland).
  - There's a **risk of misclassifying** businesses as ocean-related or not, unless you have detailed data or direct survey responses.

🔧 How It Works

- **Use existing data or projects**
  - You start with what's already available inside the country—like data from fisheries departments, port authorities, marine tourism boards, shipping registries, or past economic reports.

- **Supplement with targeted surveys**
  - Because existing data often miss important sectors, you run surveys to gather new information, especially on:
    - Employment
    - Revenues
    - Investments
    - Ocean-dependence

Different partials

Most critical component of the satellite account and must match OESA approach / method.

Level 1: Output Based

- Partial based on share of industry value added)

Level 2: Final Demand Based

- One Partial for C-I-G-X_N For Each Industry i.

Level 3: Product Based

- One Partial For Each Product for Each i.

GOAP

Estimating partials

By definition (internal)

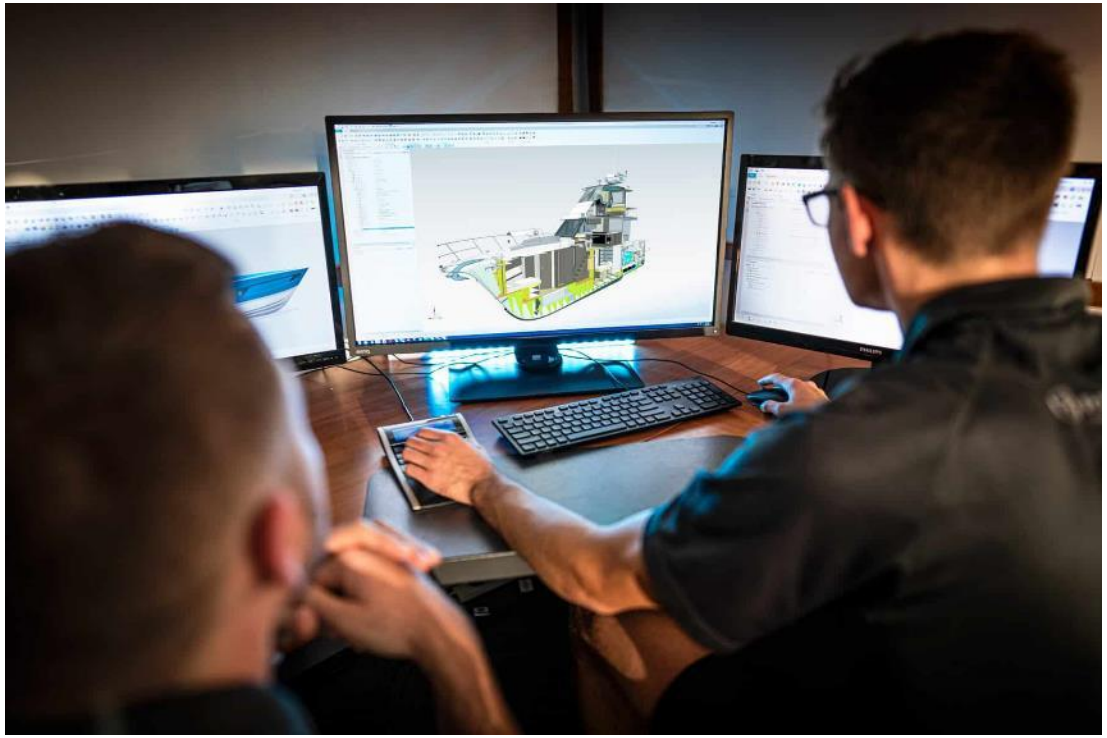
- Industries: Fishing, Marine Transport (100%, 1.0)
- Geography: Coastal Location

Ratios from data

- Any relevant data / studies.

Some partials are "obvious," but many require data, assumptions, and triangulation

Boat building (activity)



Non-ocean boats, design, components?

Not everything in an "ocean-sounding" industry is ocean

GOAP

Identifying Ocean industries and establishments

1. Industry Classification
2. Geography
   - Coastal State
   - Shore Adjacent County
   - Shore Adjacent Zip Code

Activity

Belize Classification of All Economic Activities (BCEA)

→ *derived from ISIC v. 4*

GOAP

Activity
Ocean Economy: Initial Screening of Industry Classifications

1. **Review ISIC-coded industries** provided in the activity table.
2. For each, **classify** as:
   - ***Fully Ocean*** *– 100% ocean by definition (e.g., marine fishing)*
   - ***Partially Ocean*** *– Some subcomponents use ocean resources (e.g., hotels, restaurants, tourism)*
   - ***Not Ocean*** *– No link to ocean space, services, or products*
3. Use the provided criteria to support your decisions:
   - Uses ocean space/resources
   - Occurs on ocean
   - Ocean-intended product
   - Ocean interaction

→ Share 1–2 "partially ocean" cases to compare approaches and build consensus

Partials (US OESA)

Activity	Item Description	Ocean Portion
Boat Building and Repair	Boat building - not specified by kind	33% of sales to consumers, government, and private investment
Outboard motorboats, including commercial and military (except sailboats and lifeboats)		33% of sales to consumers and private investment
Inboard cabin cruisers, including commercial and military (except sailboats and lifeboats)		70% of sales to consumers and private investment; 100% of purchases by government
Inboard motorboats, including commercial & military (except sailboats & lifeboats)		33% of sales to consumers
Inboard-outdrive runabouts, fish boats, & other inboard-outdrive boats		33% of sales to consumers
Inboard-outdrive boats, including commercial & military (except sailboat & lifeboat) - not specified by kind		97% of sales to consumers
Canoes and all other boats (excluding hovercrafts) - not elsewhere classified		33% of sales to consumers
Hovercraft		100% of private investment
Sailboats - all sizes (excluding military and commercial)		65% of sales to consumers and private investment
Boat repair, recreational		33% of sales to consumers

GOAP

Summary guide

	Data Reported	Industry Coverage	Partials	Data Sources	Price Adjustments
Level 1	Value Added	Fishing-Transportation	1.0	NA, Marine and Transport Ministries	Implicit Price Deflator
Level 2	Value Added	Add Tourism & Recreation	Geographic location	NA, VAT, Other	Price Indexes
Level 3	Value Added	Any	Any Data	NA, Other	Producer/ Wholesale/ Consumer PI
Level 4	Value Added, Gross Output	Any	VA, C, I, G, X	NA, Other	Chain Weighted Index

Examples of OESA

Countries with Ocean Economy Satellite Accounts

- USA
- Ireland
- Portugal
- South Korea
- Tanzania

Many countries have **indicators** rather than accounts.

Australia, Canada, Japan, etc.

ISIC Based Account for Tanzania

	Coast			National	Total	
	Indian	Lake	Coast Total		TZ$	Dollars
Living Resources	2,268.50	3,594.20	5,862.70	7,845.10	13,707.80	$5,914
Manufacture of prepared meals and dishes	2,041.30	1,510.20	3,551.50	1,941.50	5,493.00	$2,370
Manufacture of vegetable and animal oils and fats				5,645.60	5,645.60	$2,436
Processing and preserving of fish, crustaceans and molluscs	227.2	2,084.00	2,311.20	258	2,569.20	$1,108
Marine Transportation	4,525.30	887.6	5,412.90	5,914.00	11,327.00	$4,886
Sea and coastal freight water transport	619.3	-	619.3	3,961.40	4,580.70	$1,976
Sea and coastal passenger water transport	3,489.30	887.6	4,376.90	216.7	4,593.70	$1,982
Service activities incidental to water transportation	416.7	-	416.7	1,735.90	2,152.60	$929
Ship & Boat Building & Repair	4,401.80	3,597.90	7,999.70	139.4	8,139.10	$3,511
Building of ships and floating structures	4,396.20		4,396.20		4,396.20	$1,897
Repair of transport equipment, except motor vehicles	5.6	3,597.90	3,603.50	139.4	3,742.90	$1,615
Tourism & Recreation	238,691.30	19,248.30			257,939.50	$111,275
Beverage serving activities	100,728.10				100,728.10	$43,454
Other accommodation	36,274.60	1,447.70			37,722.40	$16,273
Other amusement and recreation activities n.e.c.	0.9	318.3			319.2	$138
Restaurants and mobile food service activities	45,342.10	3,549.70			48,891.80	$21,092
Short term accommodation activities	56,345.50	13,932.60			70,278.10	$30,318
Grand Total	249,886.90	27,328.00	277,214.86	13,898.50	291,113.40	$125,586

GOAP

U.S. Marine Economy Satellite Account By Activity

Living resources, marine
Commercial harvest, seafood markets, and processin
Commercial harvest and seafood markets
Seafood processing
Fish-based animal foods
Pharmaceuticals, marine-based
Construction, coastal and marine
Conservation
Dredging
Recreation facilities
Research and education, marine
Scientific research
National defense research and development (R&D)
Federal nondefense R&D
State and local R&D
Nonacademic R&D
Educational programs and courses
Vocational training
Laboratories

Transportation and warehousing, marine
Freight transportation
Passenger transportation
Warehousing and storage
Professional and technical services, marine
Minerals, offshore
Oil and gas
Sand and gravel
Support services
Utilities, coastal
Traditional power generation
Ship and boat building, nonrecreational
Ship building[1]
Barges and other nonpropelled ships
Military ships
Other ships
Boat building[2]
Fishing boats
Tugboats and towboats
Outboard motorboats
Inboard motorboats
Other boats

Tourism and recreation, coastal and offshore
Guided tours
Water guided tours
Other scenic tours
Recreational fishing, offshore
Boating and paddling, offshore
Sailing
Motorboating
Canoeing
Kayaking
Other boating and paddling[3]
Other water activities[4]
Other coastal recreation
Maritime museums and cultural institutions
Beachgoing
Amusement parks
Hiking and camping
RVing
Photography
Other general expenses
Trips and travel, coastal
Eating and drinking places
Hotel and lodging places
Travel arrangement services
Transportation services
National defense and public administration
National defense and coast guard
Federal public administration
State and local public administration

MESA BY INDUSTRY

Agriculture, forestry, fishing, and hunting
Farms
Forestry, fishing, and related activities
Mining
Oil and gas extraction
Mining, except oil and gas
Support activities for mining
Utilities
Construction
Manufacturing
Durable goods
Wood products
Nonmetallic mineral products
Primary metals
Fabricated metal products
Machinery
Computer and electronic products
Electrical equipment, appliances, and componen
Motor vehicles, bodies and trailers, and parts
Other transportation equipment
Furniture and related products
Miscellaneous manufacturing
Nondurable goods
Food and beverage and tobacco products
Textile mills and textile product mills
Apparel and leather and allied products
Paper products
Printing and related support activities
Petroleum and coal products
Chemical products
Plastics and rubber products

Wholesale trade
Retail trade
Motor vehicle and parts dealers
Food and beverage stores
General merchandise stores
Other retail
Transportation and warehousing
Air transportation
Rail transportation
Water transportation
Truck transportation
Transit and ground passenger transportation
Pipeline transportation
Other transportation and support activities
Warehousing and storage
Information
Publishing industries, except internet (includes softw
Motion picture and sound recording industries
Broadcasting and telecommunications
Data processing, internet publishing, and other info
Finance, insurance, real estate, rental, and leasing
Finance and insurance
Federal Reserve banks, credit intermediation, an
Securities, commodity contracts, and investments
Insurance carriers and related activities
Funds, trusts, and other financial vehicles
Real estate and rental and leasing
Real estate
Housing
Other real estate
Rental and leasing services and lessors of intang

Professional and business services
Professional, scientific, and technical services
Legal services
Computer systems design and related services
Miscellaneous professional, scientific, and techni
Management of companies and enterprises
Administrative and waste management services
Administrative and support services
Waste management and remediation services
Educational services, health care, and social assis
Educational services
Health care and social assistance
Ambulatory health care services
Hospitals
Nursing and residential care facilities
Social assistance
Arts, entertainment, recreation, accommodation, an
Arts, entertainment, and recreation
Performing arts, spectator sports, museums, and
Amusements, gambling, and recreation industries
Accommodation and food services
Accommodation
Food services and drinking places
Other services, except government
Government
Federal
General government
National defense
Nondefense
Government enterprises
State and local
General government
Government enterprises

GOAP

Choice of method

- **Data Availability and Institutional Capacity**
  - If a country has **detailed establishment-level data** (business registers, surveys by sub-industry, regional data) that can identify ocean activities, a bottom-up approach can be very effective and may be easier to execute by a small expert team.
  - if a country has a well-developed system of **national accounts with input-output or supply-use tables**, it might leverage that for a top-down account
  - each country chooses a strategy based on its data strengths

Global Ocean Accounts Partnership

Thanks for listening!

GOAP